

Manager Dashboard Overview - Change Management System

This overview explains how to visualize Change Requests, Resource Utilization, and Task Conflicts using the new `cm\_` schema and unified Supabase views integrated with your Lovable SaaS Event Planner.

1. Dashboard Purpose

The Manager Dashboard consolidates operational visibility across event changes, resource constraints, and task dependencies. It combines data from `cm\_change\_requests`, `cm\_tasks`, `cm\_resources`, and `cm\_audit\_events` through the unified Supabase views (`unified\_\*`).

2. Unified Data Sources

The following unified views were created to merge main and cm_ datasets for analytics:

- **unified_tasks** – Merges live and change-managed task timelines.
- **unified_resources** – Displays cross-venue resource allocation and availability.
- **unified_audit_events** – Tracks all change requests, timeline adjustments, and resource updates.
- **unified_locations** – Provides the contextual hierarchy for multi-location planning.

3. Recommended Dashboard Layout

A. Change Request Insights

- Bar or pie chart: Requests by Priority Tag (Urgent, Optional, Deferred).
- Line chart: Change Requests per Day/Week (trend analysis).
- Table: Latest requests with user name, description, and approval status.

B. Task Management Metrics

- Gantt timeline for task start/end with color-coded status.
- Conflict heatmap: Tasks overlapping with locked deadlines.
- Resync history log via unified_audit_events (type='timeline_resync').

C. Resource Utilization

- Table or bar chart: Resources by role and location.
- Availability trend: percentage of booked vs free slots per venue.
- Vendor/service cost tracking via custom resource tags.

4. Supabase SQL Queries for Dashboard Data

These sample SQL queries feed the dashboard widgets:

Change Request Counts by Priority

```
SELECT priority_tag, COUNT(*) FROM unified_change_requests GROUP BY priority_tag;
```

Timeline Conflict Summary

```
SELECT event_id, COUNT(*) AS conflict_count FROM unified_tasks WHERE locked = TRUE GROUP
```

BY event_id;

Resource Utilization

SELECT role, COUNT(*) FILTER (WHERE availability IS NOT NULL) AS assigned, COUNT(*) FILTER (WHERE availability IS NULL) AS unassigned FROM unified_resources GROUP BY role;

Audit Event Feed

SELECT type, description, created_at FROM unified_audit_events ORDER BY created_at

DESC; **5. Visual Integration (Lovable UI or BI Tools)**

You can embed visual widgets directly into your Lovable dashboard using the built-in chart blocks or connect your Supabase project to BI tools like Metabase, Retool, or Grafana. Use `unified\_` views for data sources to maintain parity between your main and cm_ environments.

6. Key Metrics for Managers

- Number of change requests submitted this week
- Percentage of Urgent vs Optional requests
- Number of locked tasks vs total tasks
- Number of timeline resync events
- Resource utilization rate (per role or location)
- Average change request resolution time

7. Alerts and Thresholds

Configure triggers or dashboard alerts using Supabase Functions or BI rules:

- Alert when Urgent requests > 5 per event
- Alert when resource utilization > 90%
- Notify when any locked task is modified
- Daily digest email: summarize new change requests and audit logs

8. Future Enhancements

- Add predictive analytics for resource forecasting (using `availability` JSON data).
- Integrate Slack or Email notifications for timeline resync events.
- Extend unified_audit_events to capture user device and location metadata.
- Integrate manager approval flow for Urgent requests.

9. Summary

The Manager Dashboard centralizes all key operational metrics, ensuring every event adjustment is visible, auditable, and measurable. The use of `unified\_` views bridges live and change-managed data, providing a single truth for all event operations.